Plant/department:	RzP
Project:	Insourcing solenoid armature
Date:	02.04.2024
Revision Nr.:	1
Machine or equipment type:	Assembly line or round table
Time plan:	
Quotation delivery: to TME or PUR	01/05/2024
Confirmation of specification to Project manager	15/04/2024
Pre-acceptance: (at supplier):	15/02/2025
Delivery no later than:	01/03/2025
Start of Production no later than:	10/04/2025
Final acceptance: (at operation location)	15/03/2025

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Date; signature	xx.xx.xxxx; _____

With the signature supplier confirms the adherence to the specification. Additionally supplier should fill out the table of “manufacturer response to spec”

Date	Signature of supplier

Project versioning

Version	Dat.	Proposed by	capital	Change
1	02.04.24	MFE1Ba		Version for first quote
2				
3				
4				
5				
6				
7				
8				
9				
10				

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A. Project specific technical specification

The specification describes the project requirements for building up the customized assembly machine/line machine of certain application.

Based on the technical requirement, the supplier is expected to develop a viable solution/concept to meet the technical requirement. This should be submitted in a formal quotation to the requesting buyer.

Supplier is requested to develop and implement innovative/robust/lean process and machine concept for Bosch projects. In the event this leads to deviation from the technical specification, written agreement with RB project leader is necessary.

Bosch Norms and guidelines can be found in the download area via link:

Delivery instructions for machinery

or can be provided by the Bosch project leader.

A.1 Task of the equipment/ machine/production line

This project is about the design and the implementation of an automatic equipment to assembly the solenoid armature. It is a sub-assembly of the PCV (pressure control valve) which is a component of the injection chain.

Solenoïd Armature is made of two components which are assembled in force:

The armature (in green) and the pin (in red) will be manufactured in Rodez plant too.

This equipment must assembly the two parts according precises instructions of forces and the result must be in conformity with precises geometry in term of axial and radial runout and straightness.

An identification of each plate will be implemented by a data matrix code

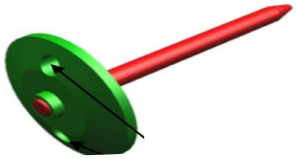
There are 4 different references made from 3 plates references and 3 pin references.

A device to avoid to mix each plate reference and each pin reference must be implemented.

Only pins will be heat treated. Plates will arrive in this equipment after turning and without heat treatment.

No damage and no burrs are allowed on the assembled parts. Surface of the plate is particularly sensible and important.

The assembly needs the control of some characteristics (length position, radial and axial runout, press-fit effort etc..) and some of them have to be saved.



Change over must be made with a maximum time of 20 minutes.

After 3 bad parts consecutively in the assembly operation, the equipment must stop and display an alarm.

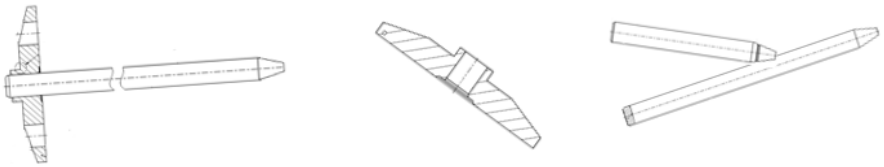
An early warning system must be provided to stop the line when X% of scrap parts are reached. This percentage will be the variation in type data. For reasons of cycle time, several operations can be carried out in the same station.

A.2 Description of product and manufacturing requirements

A.2.1 Product variants, drawings

A.2.1.1 Part number

Parts name	Assembled part reference	Plate reference	Pin reference
Solenoid armature	F 00N 000 182	F 00V D43 207	F 00N 000 183
	F 00N 000 375	F 00N 000 379	F 00N 000 274
	F 00V D43 203	F 00V D43 200	F 00V D44 200
	F 00V D43 208	F 00V D43 207	F 00V D44 200



A.2.1.2 Manufacturing and testing instruction.

production and testing instruction	Valid date
F 00V D39 030	23.06.2008
F 00V D39 031	09.06.2008

The production and testing instruction are attached.

A.2.2 Agreed Characteristics Covenant

A.2.2.1 performance record

Performance criteria	required	Unit
Machine cycle time	< 8.2	s
Total NOK parts	2	%
Number of operators, machine supervisor	0.5	ETP
Technical availability	>85	%
Change-over time (2 operators)	<30	Min
		%
Machine capability (50 parts)	>1,67	
Measurement device capability for the quality relevant features	CGK > 1,33 GRR% <10	

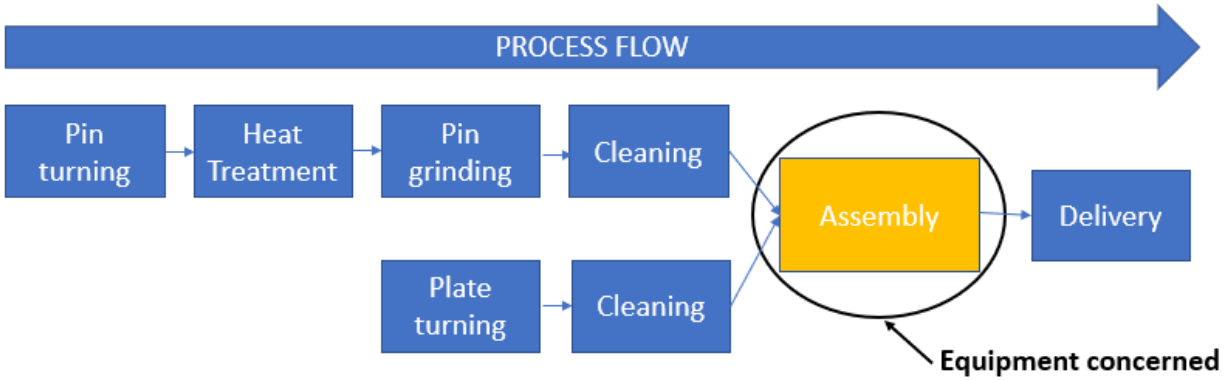
For the critical stations the CT analysis or simulation is necessary and is to be presented to project leader.

A.2.3 Layout

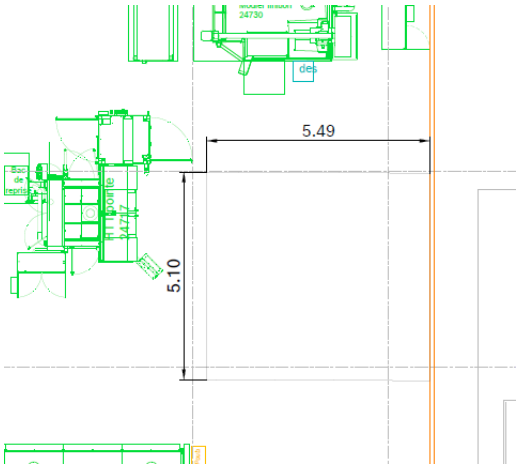
Following aspects are to be considered for the machine concept design:

Characteristics	Value
Color of the equipment	RAL 7035
Color of the control cabinets	RAL 7035
Setting up area	Shopfloor

Room height	3m (under canalis)
Floor strength	tbd
Max. transport dimension (i.e. for lift, transport truck, air freight, transport route in the plant)	tbd
Max. transport weight (i.e. for lift, transport truck, air freight, transport route in the plant)	tbd



Assembly machine will be put in the area below:



If this area is not sufficient, Bosch project leader must be informed as soon as possible.

The machine layout can be prepared in the format of.dwg or .dxf and is a part of the design meeting with Bosch. The final machine or line layout should be released by the project leader.

A.3 Description of the assembly sequence and process requirement

This equipment will have several functions. Its design must take in consideration Bosch need an evolutive machine and want to keep the possibility to add or remove operations easily.

Below, are described the stations we currently need:

A.3.1 Pin trays loading

Pins will be delivered in trays (see drawing ET-0746W004 for diameter 4 and … for diameter 5). We will receive 3 classes of parts according to the diameter.

References with diameter 4 mm:

Class 1: 3.993 - 3.995

Class 2 :3.995 – 3.997

Class 3: 3.997 – 3.999

References with diameter 5 mm:

Class 1: 4.993 - 4.995

Class 2: 4.995 – 4.997

Class 3: 4.997 – 4.999

Limits of each class must be configurable.

This station will be designed to contain 1 tray (ET_0746W004) of each class in a single reference.

When a tray is empty, machine must display an alarm and a light must flash.

We have 3 pin references (see paragraph A.2.1) in annexed drawings.

- Tray drawings are in appendices

A.3.2 Plate loading

Plates will be delivered in trays which are not yet defined.

The correct positioning of the part must be guaranteed in the machine.

3 plates references are possible (see paragraph A.2.1 and annexed drawings).

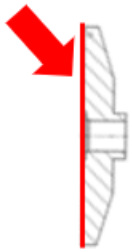
Autonomy: Machine must contain almost 4 trays or a time of autonomy of 30 minutes

It must be possible to change the empty tray by a full one without stop the machine and in a safe condition. (Possibility to use continued loader system with a conveyor belt)

A.3.3 Plate brushing

Each plate must be brushed to avoid burrs around the holes and the external diameter.

The surface which must be brushed is indicated below:



No burrs are allowed after brushing and the brushing mustn't damage the plate surface or do marks on this surface.

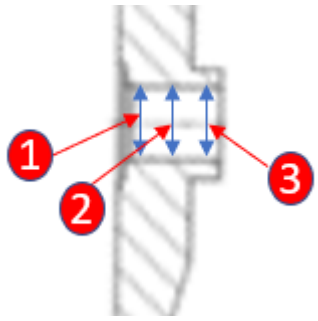
If it's needed, an automatic compensation of the wear of the brush must be implemented. The length life of the brush expected is almost 10 000 parts. A specific counter must indicate the quantity of parts done by the brush.

After brushing, parts must be blown to avoid issues in the following steps.

This operation must be carried out in a closed housing (bell jar) to avoid pollution in the other stations with particles. This housing will be connected to the exhaust network of the building.

A.3.4 Hole plate measurement

The hole of the plate must be measured by an accurate air flow device. It must be possible to visualize the form of the hole and the values of the diameter in 3 positions (beginning, middle and end). This 3 values must be recorded too. If one of the value is out of limit (configurable), part must be scrapped.



To be sure of the good running of this device in the time, a stability part must be automatically measured. It must be possible to change the frequency of this measurement. It will be also possible to check the calibration by a request of the operator.

A calculation of the delta between the max diameter and the min diameter must be made. If this result is more than a limit (configurable), part must be scrapped.

This measuring devices will be released according to the CGK and GRR % methods

The stability measurement will be made like it:

- 3 measurements of the stability parts with recording of the middle value (2)
- Calculation of the mean value and the standard deviation and recording of these values
- Comparison to limit values.
- Limit values can be editable by password.
- If the values are not between the limits, machine must stop and put a message.
- It must be impossible to start again the machine in production if these values are not inside the limits.

Calibration: The calibration must be made in a diameter (middle diameter). It can be made manually with a standard part.

A.3.5 Assembly device loading

According to the diameter of the hole of the plate, the machine must choose a class of pin and load a part from the respective tray to the pin fixture under the press.

Next, the machine must take the plate, load it to the plate fixture under the press

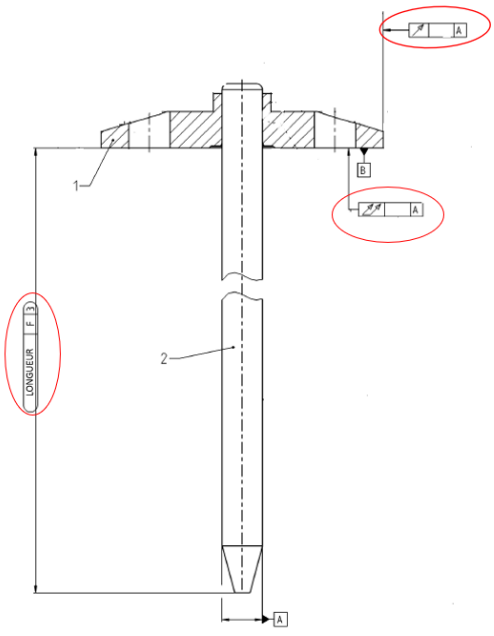
The algorithm to choose a pin according to the diameters of the plate must be defined and configurable.

A.3.6 Press fitting

The two parts of the solenoid armature must be assembled using a press.

Appropriate tools must be provided to ensure that the assembly according to the following requirements:

- assembly fixture
- radial pin runout in relation to plate
- double axial pin runout in relation to the plate
- press-fit effort
- position Pin/plate



The assembly operation must not cause any damage, any scratches on the two parts.
The straightness of the part mustn't be increase than more 2µm after fitting.
Permissible press-fit forces are specified on each drawing (supplied as an attachment).
It must be checked in 4 areas in process.

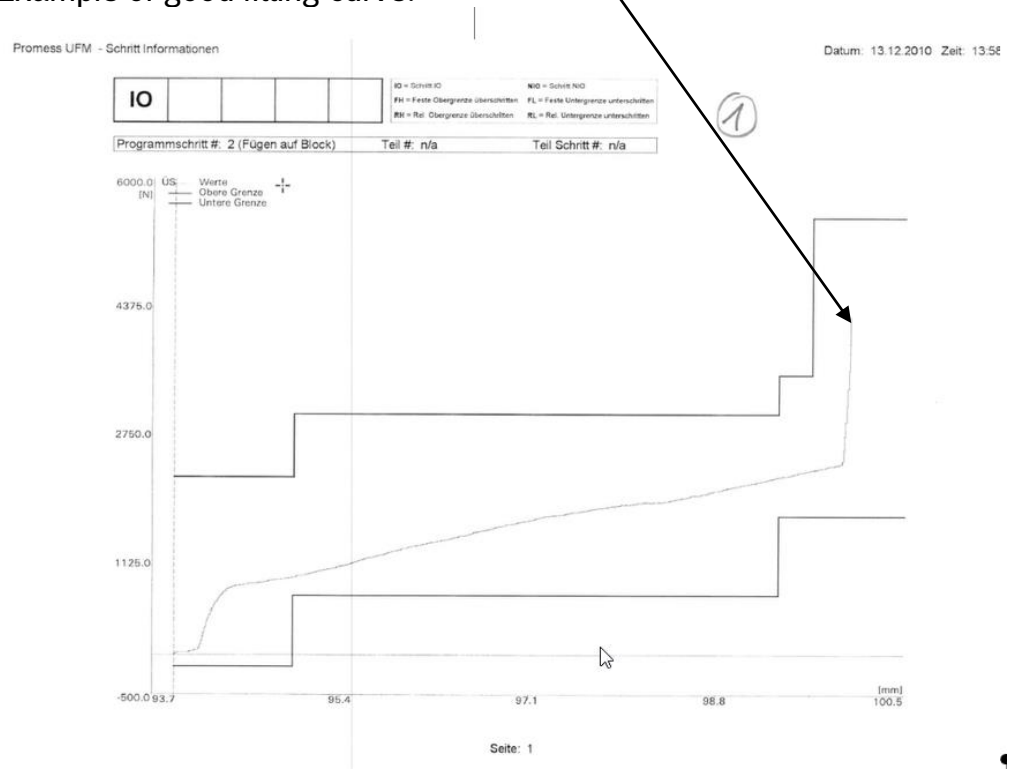
Example:

press-on (see specification pos. 970.0) Aufpressen (siehe PV - Pos. 970.0)			
	min. force in kN min. Kraft in kN	max. force in kN max. Kraft in kN	remark Bemerkung
s ≤ -4.6	0	2.25	
-4.6 < s ≤ -0.6	0.75 (02)	(4) (3.0 F 3)	
-0.6 < s ≤ -0.3	1.75	3.5	
-0.3 < s ≤ 0	1.75	5.5	

further press-on (see specification pos. 970.0) Weiterpressen (siehe PV - Pos. 970.0)			
sort of force Kraftart	point Ablesepunkt	required value Sollwert	unit Einheit
further press-on force absolute Weiterpresskraft absolut	F-max.	min. 2	kN
further press-on force relative Weiterpresskraft relativ	F-max.	min. +0.3	kN

0 is the end of the stroke. (Stop position)

Example of good fitting curve:



. Each force value of the area -4,6<s<-0,6 must be recorded. Curves must be displayed in a screen during the process.
If one of the efforts is not correct, the press must stop its stroke and set on an alarm.
Restart with an acknowledgement, and the part will be scrapped.

The measurement of the effort will be regularly over checked by our quality department with a sensor Kistler. So, the fitting device must have the possibility to integrate the sensor “Piezoelektrischer Kraftaufnehmer, Kraftmessdose (Fz ±24 kN) / 9331C” below :

https://www.kistler.com/DE/de/p/piezoelektrischer-kraftaufnehmer-9331c/000000000018037812



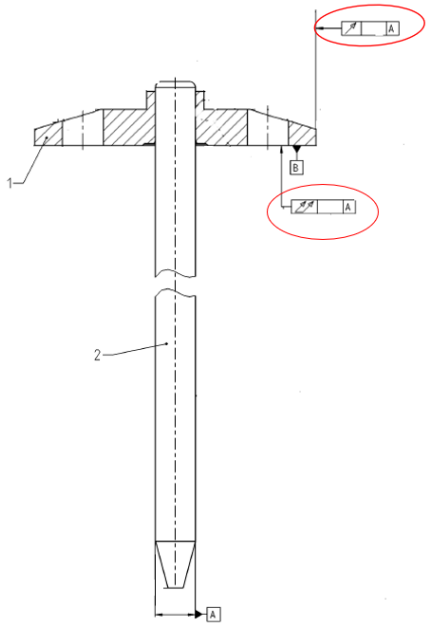
Parts and programs to allow to overcheck the effort measurement of the machine with the Kistler sensor must be defined and done by the supplier of the machine.

Fitting operation must be done according to the documents F00VD39030 and F00VD39031

A.3.7 runout checking

The axial and radial runout of the armature to the pin axis must be 100% checked. Required accuracy: 1 µm.

These two measuring devices will be released using the CGK and GRR % methods.



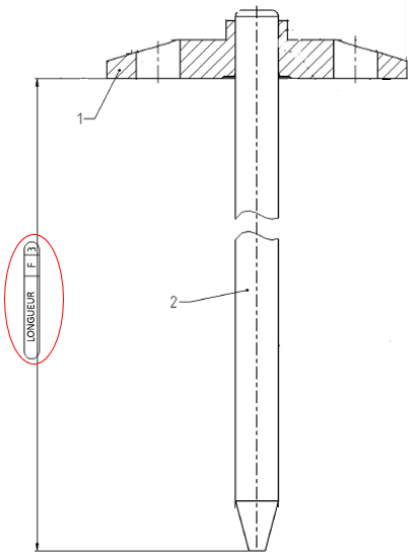
Permissible tolerances are specified in drawings F00V D43 203, F00N 000 375, F00N 000 182 and F00V D43 208 (in appendices).

If any of the values measured are outside tolerance, the part will be scrapped.

A.3.8 length checking

The position of the plate to the pin must be 100% checked by a measurement device. Required accuracy: 1 µm.

Release of this measurement device will be released using the CGK and GRR % methods.



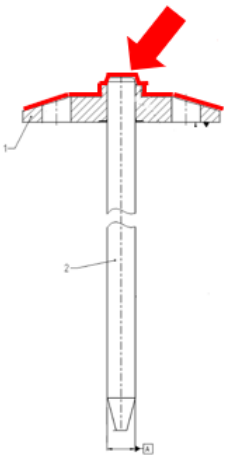
Permissible lengths are specified in the drawings F00V D43 208, F00N 000 375, F00N 000 182 and F00V D43 203 (in appendices).

Nb: if this is technically possible, the 4.3.1.4 and 4.3.1.5 can be done in the same station

Calibration must be made in automatic when the operator requests it.

A.3.9 Back brushing

A brushing operation on the surface of the solenoid is required to remove burrs generated by press fitting and turning operation.



Burrs are not allowed after brushing.

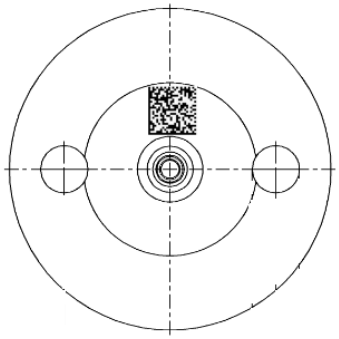
If it's needed, an automatic compensation of the wear of the brush must be implemented. The length life of the brush expected is almost 10 000 parts. A specific counter must indicate the quantity of parts done by the brush.

After brushing, parts must be blown to avoid issues in the following steps.

This operation must be carried out in a closed housing (bell jar) to avoid pollution in the other stations with particles. This housing will be connected to the exhaust network of the building.

A.3.10 DMC Marking (in option)

Parts will be identified by a Data Matrix Code. This laser marking must be made in the back of the part.



Be careful: there are additional holes on several references. Marking must be fully out of these holes.

A camera will check the good marking of the part.

The goal of this identification is to have a traceability of the part. It must link to:

- The reference of the part
- The number of the lathe

- The number of the batch heat treatment of the pin
- The assembly data's
- The assembly date

To reduce the costs, il will be possible to reuse a Laser marker supplied by Bosch with following characteristics:

Brand name: Trumpf, model : Trumark 6000, type : VMC3-class4, number : 0562180

A camera could be reuse too : Brand name : Cognex, Serial n° : Z62979133, Part number: 800-5870-1

A.3.11 Free station

In case of a need in the future

A.3.12 Unloading

The assembled Solenoid Armatures will be put in 50 parts trays (supplied by Bosch RzP). Only complete trays (without empty positions) will leave the machine.

Tray reference: ET0830W001

Autonomy: Machine will contain at least 4 trays

4 reject boxes will also be positioned near the tray but will be inaccessible to the operator in automatic mode. They must be with red color.

Scrap box 1: press-fit effort reject.

Scrap box 2: runout reject.

Scrap box 3: press-fit position reject.

Scrap box 4: Internal diameter of plates reject.

Each scrap box must have a capacity of 20 parts at least.

A.3.13 Further press on

A function will allow to do the operation "Further press on". This operation consists in press more and measure the effort. Specifications are mentioned in the drawings.

Example:

further press-on (see specification pos. 970.1) Weiterpressen (siehe PV - Pos. 970.1)			
sort of force Kraftart	point Ablesepunkt	required value Sollwert	unit Einheit
further press-on force absolute Weiterpresskraft absolut	F-max.	min. 2	kN
further press-on force relative Weiterpresskraft relativ	F-max.	min. +0.3	kN

Specification to do this operation are described in the documents F00V D39 030 and F00V D39 031. A curve of the force-stroke in appropriate scale must be recorded.

Appropriate tools to do this operation must be supplied. The change over must last less than 10 minutes.

A.3.14 Summary of all commissioning components and subjects

The provided components, services and know-how (process descriptions) are Bosch property and may only be used for the project discussed here. These are subject to Bosch IP-right. These can only be used for the Bosch project defined in this specification.

A.3.15 Summary of configurable values for stability and calibration:

For each measurement station (diameter, runout 1 and 2, length) we will have parameters below:

	Configurable values		modification	Display	Record
Calibration	Calibration part value *		Password	Yes	No
	Calibration interval *		Password	Yes	No
Stability	Stability checking interval		Password	Yes	No
	Stability part value		Password	Yes	No
	Measurement 1			Yes	Yes
	Measurement 2			Yes	Yes
	Measurement 3			Yes	Yes
	measurement mean	calculated value		Yes	Yes
		max limit	Password	Yes	No
		min limit	Password	Yes	No
	Standard deviation	calculated value		Yes	Yes
		max limit	Password	Yes	No
		min limit	Password	Yes	No

* Only when the calibration is done in automatic

A.3.16 Stabilities running

A stability checking will be done every X part (configurable): a stability part (supplied by Bosch RzP) will be automatically loaded and measured 3 times consecutively. The average and standard deviation of these 3 values will be calculated by the machine and recorded. If these values is between the limits (configurable), the machine will restart in production. If not, it will stop and an alarm will be displayed.

The request of stabilities and calibrations will be possible station by station.

We will have a game of calibration and stabilities parts for diameter 4 mm references (F00N000182, F00N000375, F00VD43203, F00VD43208) and another game for diameter 5 mm reference (F00N000375).

No need to store calibration and stability parts for types not used on the machine.

A.3.17 Stability and calibration mode summary:

Station:		Plate internal diameter	length fitting	radial runout	double axial runout
Calibration	Manual	x		not concerned	not concerned
	Automatic		x		
Stability	Manual				
	Automatic	x	x	x	x

When calibration and stabilities are automatic, it must also be possible to request it by the operator.

A.3.18 Station selection

Each station can be activated or disabled by selection from the desk (locking by password). The machine must be able to run with a disabled station.

A.4 Change-over concept

A.4.1 Start of change-over at beginning of the line (please choose)

To be discussed according to the concept proposed

- ☒ manually performed at each station
- ☒ by the selection of type number at the first machine of the line
- ☐ by scanning of a Kanban card with Barcode of type number (number of parts)
- ☐ by the activation with a MES functionality (MES-specification) for all stations
- ☐ others

A.4.2 Method to Empty Line

☐ Separate function to empty the machine is not necessary.

☒ Start of empty the machine is activated by:

- ☒ Change-over
- ☐ Activation of a separate function at each station

☐ The line has to be emptied in order to separate the different types.

☒ The emptying is initiated at the first station.

☐ Necessary number of empty WPC for the production line

☒ During the changeover/emptying, the feeding systems / bunkers / shelves empty to prevent the mixing of component types at the station.

☒ NOK-Parts must be removed manually by operator from the shelves/boxes/ slides.
If is necessary:

- ☐ the emptying is monitored by sensor or
- ☐ the emptying is controlled by HMI measures

A.4.3 Change-over time loss

Target: <30min

☐ A changeover gap must to be avoided

☐ The changeover gap must be limited to a maximum of ___– machine cycles.

☐ The changeover gap must be limited to a minimum by supplier.

☒ The size of the changeover gap must be proven by the supplier and explained in the concept review.

A.4.4 Change-over methods

☐ Fully automatic setup

- ☐ The machine should transfer the necessary data and the mechanical setup can be performed independently
- ☐ The machine recognizes the type of number or changeover marker by
  - ☐ reading of an identification system (for example ID200)
  - ☐ request of the Part-IDs of the connected MES (MES specification)

☒ Semi-automatic setup

The machine / line can partially automatically exchange tools and / or goes to a position where the conversion of e.g., Tools can be done by the operator.

☐ Manuel setup

The operator has the option to convert the system manually. That in a special operating mode, he can select the setup for another TTNR, which is to be equipped and is guided through the setup process by operator instructions.
Components, tools, grippers, brackets, nests, etc. must be replaced manually.

- ☐ In the display of the machine the tool to be exchanged, the holder, unit visualized and / or plausibility and / or sensory monitored or
- ☐ Correct installation of tools, grippers, holders, nests, etc is to be guaranteed by (for example Poka Yoke, ...

A.4.5 Production of types in mixed operation

☒ The production of a new type (and variants) must be carried out by a setup software.

☐ Different types (and variants) of a product are.

- ☐ producible without further mechanical intervention
- ☐ only producible by another mechanical intervention

A.4.6 Change-over data

The needed change-over data are:

☒ To keep local (i.e., on the control), to document and to prove a backup concept of the setup data when the system is delivered.

☐ Central (i.e., from an MES or SCADA) (see MES specification). A corresponding data backup concept of the setup data must be verified. The setup data must be reproducible at all times.

☐ If setup data is to be obtained from an identification system (e.g. ID200), this must be documented and verified when the system is delivered in the operating instructions.

A.5 Scalable concept

Not concerned.

A.6 Processing concept for faults / NOK parts

A.6.1 List of irreversible and reversible processes

No.	Process	Description	Arrangements by NOK
1	Plate Diameter	irreversible	scrap
2	Fitting	irreversible	scrap
3	Brushing	reversible	Manual 100% checking of the parts and new brushing if needed
4	Laser marking	irreversible	scrap

A.6.2 Description of the parts after processing

Not concerned

☐ Re-worked components are to be introduced at separate points in the line. (Add description / concept)

☐ Reworked components must be self-contained until further process be passed through

☐ Refinished components are visualized by the display of the system

☐ Refinished components must be documented and described in the MES specification

- ☐ Disassembled A components must be documented and scrapped. Assembled A components have to be detected
- ☐ Assembled B components have to be detected

A.6.3 Software based NOK-part processing

☐ As the rework is a pure / complementary software-based rework (Data Maintenance) and is described as a rework concept in the MES specification.
The rework is done at a rework station with a PC and needed

☐ ID identification systems. The components are thereby on the required process coded.
Required PC's and ID identification systems must be offered

☐ Required specification of the PC's and ID identification systems have to be described and procured by the operator

A.6.4 Reworking with own processes

☐ The required workstation and ID identification systems have to be offered. At the Reworking work is carried out dismantling up to the previously valid IO Process

- ☐ The rework station has all necessary means to be able to dismount something
- ☐ The rework station has all the necessary means to carry out an assembly / measurement up to a required process
- ☐ Data maintenance takes place through the rework station and is described in the MES specification. The components are thereby coded to the required process

☐ After reworking, the component is fed through the rework station into the production line

☐ The required rework takes place in the production line at the process (description, reference to list)

A.7 Data management and communication with manufacturing execution system, MES

The standard manufacturing execution system of Bosch is Nexeed MES. Its communication with the MACHINE and stations is enabled by a data collection.

A.7.1 Data classification

Data to be collected and processed can be defined principally with 9 classes:

- machine data
- process data
- type data
- tool data
- material data
- version and NOK concept for each type
- condition monitoring
- energy monitoring

- required additional data, which are necessary for the project or plant.

A.7.2 Overview of data requirement

N°	Line	Process N°	Description	ID Bulder	Process data	Machine data	Quality Gate	Type Data	Rework
1		tbd	Plate Brusching		x	x		x	
2		tbd	Hole plate measurement		x	x	x	x	
3		tbd	Press on fitting		x	x	x	x	
4		tbd	Bach brushing assembled part		x	x		x	
5		tbd	Length checking		x	x	x	x	
6		tbd	Double Axial runout checking		x	x	x	x	
7		tbd	Radial checking		x	x	x	x	
8		tbd	Data Matrix Code marking		x	x		x	
8		tbd	Unloading		x	x		x	
			TOTAL		9	9	5	9	0

A.7.3 Detailed data requirement

This point will be discuss with the supplier.

A.8 Acceptance

A.8.1 Design acceptance before procurement of the components

The design acceptance have to be performed at the time of the completion of the design (concept) for stations that have been modified or further developed for presentation. The goal is the conceptual approval of the design including the mechanical and electrical components used to ensure the standards for manufacturing processes, assembly steps and station construction. For this design acceptance, the contractor agrees with the project manager on a joint appointment, preferably via online meeting.

The submission of the acceptance document for the design acceptance could be found in the appendix. CGP 80006-140 Acceptance protocol

A.8.2 EPlan and Software review (optional)

The E-Plan P8 and Software construction pre acceptance occur during the pre-acceptance of the Production line.

Therefore, the implementation of the risk assessment of the machine security must be checked in the E-Plan, the observing of BMK determination of Bosch, the observing of construction guidelines and the maintainability of the electrical components.

The software construction pre-acceptance must be checked based of the software programming instructions. At least hand functions for movement and teaching operation, the visualization and the information necessary for the operation of the plant are available. The special operation contains at least the basic position travel, the automatic operation must contain all processes necessary for the production process and the communication necessary for the connection to an MES which supports the manufacturing process and / or logged.

A.8.3 Pre Acceptance at supplier site

Before the machine is delivered to Bosch a pre-acceptance will be carried out. Place and timing are to be agreed with Bosch project leader timely

Upfront the pre-acceptance supplier is required to prepare the equipment ready for operation, which is functioning and capable to meet the agreed technical performance.

Should the pre-acceptance be repeated, due to any reasons responsible by supplier, the expenditure and effort happened to Bosch should be compensated by supplier.

The tooling and materials provided by Bosch as provisioning must be requested by supplier 8 weeks before scheduled pre-acceptance

The equipment is to be setup and operated with the parts, listed in the following table:

part-name	Drawing Nr. or part number	quantity for pre-acceptance
Solenoid Armature	F00VD43208	5000

The pre-acceptance is performed with electrical connection, correspondent to the final arrival country. The agreed performance will be investigated with 3500 pieces of products, which is carried out at supplier with the presence of Bosch. The agreed performance, like output (CT) and change over time will be measured with proper method and measurement devices.

Besides that the technical construction of the machine will be released by our technician according to the checklist and requirements defined in the technical specification. This check list CGP 80006-140 Acceptance protocol can be obtained by Bosch project leader. In the case of

significant technical deficits and deviation from the technical requirement supplier must remedy them. A new pre-acceptance must be planed again.

A.8.4 Final acceptance at Bosch plant

The final acceptance of the machine is to be carried out at Bosch after the machine is taken into operation at the final arrival location under such condition:

- ☐ Error free trial operation
- ☐ Proof of the agreed machine performance and technical availability of the machine
- ☐ The final acceptance requirement, based on BOSCH-Norm N51M M20 and all other agreed technical instruction are fully met
- ☒ The complete documents according to BOSCH-Norm N51M M20 are handed over to Bosch
- ☒ 3-day-production with one type (agreement with Bosch project leader)
- ☒ Change over and 4 hour-production of several required product types

In the frame of final acceptance, the whole delivery scope, energy consumption and other consumption will be measured and compared with the data provided by supplier. After all the criteria are fulfilled the final acceptance can be completed and the warranty time starts. The acceptance protocol will be prepared together

A.9 Personnel instruction, training and documentation

A.9.1 Personnel instruction and training

A detailed training concept it to be prepared and introduced (training content, period, location and participant number, description of the necessary qualification of the participants).

Following training packages are to be considered in the quotation:

- ☐ Basic training at Bosch location for X operators (location) for X day(s)
- ☐ Training at supplier for X operators for X day(s)
- ☒ Training at Bosch for 2 operators for 5 day(s)
- ☐ Extended training at supplier for X operators for X day(s)
- ☐ Extended training at Bosch for X operators for X day(s)
- ☐ Training Bosch service staff for warranty reparation
- ☒ System training (Programming) for 3 technician (with good knowledge) for 5 day(s)

The training concept must be, in agreement with Bosch project leader, adapted to staff, who are to be trained and adapted to the product, which is to be produced.

A.9.2 Number and languages of the documents

Location	Required language	Number of documents to be handed over	Number of data media (CD)
RzP	French	3	1
RzP	English	2	1

B General construction guideline

B.1 Machine legal regulation and Bosch Norm

The functional safety of the machine and equipment must be ensured, according to N51M M20. The basis for the requirement on the “machine safety” – safety relevant control components, is the DIN EN ISO 13849-1.

For the safety function validation of the machine and equipment with safety relevant control components (SRP/CS) the software SYSTEMA of the **I**nstitute for **O**ccupational **S**afety should be used. The SISTEMA project is to be transferred to Bosch with the complete machine delivery.

The manufacturing operation is expected to perform in the individual manufacturing station. Each station is, according to 2006/42/EG (machine guideline), a separate machine. The structure of the whole equipment should be designed in the way that no total declaration of conformity is required. Only the individual declaration of conformity for individual machines is necessary, although the machines need to be chained up with each other at last.

In case that the individual declaration of conformity is, due to technical or process reason, not possible, supplier is obligated to provide a total declaration of conformity for the chained equipment. In this case the organizational reason is not acceptable. A total risk evaluation at the interfaces and related risks, especially the followed measures against those risks must be developed and delivered to Bosch. Supplier must deliver an additional operation instruction for the whole equipment.

For integration of old machines into a new machine or machine chain, supplier must check, whether the machine structure and configuration thereafter can still fulfill the current machine guideline requirement. It is important to check, whether the old machine can still meet the requirements of the current machine guideline. If not, the old machine must be improved, so that the conformity can be fulfilled.

Despite the declaration of conformity for the whole newly chained equipment, supplier should deliver the declaration of conformity of each integrated machines.

The equipment can only be taken into operation, after the declaration of conformity is completely handed over to Bosch and CE-label is placed onto the machines.

By the application of contactless or mobile separable protection measures supplier should provide simulation or measurable proof, to demonstrate that the requirement of Norm DIN EN ISO 13855 can be fulfilled.

B.2 Mechanic

B.2.1 Additionally applicable documents, laws and standards

- N51M M20 general terms of delivery for machines, equipment and production equipment (MAE)
- N51M M23 machine tools, terms of delivery; terms of delivery for electrical, pneumatically and hydraulically equipment from MAE
- N51M M24 Delivery instructions for ESD-compliant production components
- MAE target specification Bosch TPM Pillar 4
- N62 A ergonomic

B.2.2 General instruction about the mechanical design

The machine should be designed for an operation time of 10 years, with 3-shift-operation and 18 shift /week. From the operation time and machine cycle time, the cycle number of the wear parts and life time of the components can be derived and chosen accordingly.

The preferred components can be found in the component list of “preferred standard components”. Delivery instructions for machinery. Deviation is possible, if it is aligned with and agreed by Bosch project leader.

For the design of a component or sub-assembly, which should be integrated into a machine, attention should be paid to the interfaces of adjacent machines and processes. Necessary information should be requested at the Bosch project leader.

In the connected automatic manufacturing line, the machines should be numbered along the manufacturing flow. For the manufacturing station an explicit number for each station must be ensured. The station number should be visible from outside and not lost over time.

B.2.3 Installation condition and available connections provided by the Bosch

For the installation condition the plant specific inputs are valid, which can be available through the Bosch project leader

pin	Value*
Environment temperature grey room	20-50°C
Environment temperature clean room	20-26°C
Relative humidity- clean room	30 – 60%
Power supply /USV	400 V 50 Hz, no USV available
Plant data network	Single mode fiber optic connection 1 Gbit
Compressed air dry, unoiled	6 bar (Quality)
chilled water:	18°C, pressure difference 2,5bar
Cooling water	≤26°C, pressure difference 2,5bar
Drain connection	Alignment with HSE necessary
Steam and gas extraction	Alignment with FCM/HSE necessary

*could be specific to plant.

Water carrying pipes, hoses and components, together with their connections should be designed in compatible with the installation guideline DVGW and meet the minimal requirement of pressure level PN10 (10 bar), which is based on EN1333.

Regarding internal shut-off valves, the pressure shock in the cooling water system must be avoided. The application of gradually closed shut-off valves is recommended.

The supply of process liquids to the machine should be shut on based on demand. The pressure level and amount of the process liquids should be regulated according to the demand.

Each line part has to be arranged with his own feeding table. This represent the feeding of power, air and cooling water across the provision of the hall to the line part. Preferably, this table is installed at the beginning or the end of the line.

B.2.4 Instruction to the station building

B.2.4.1 General design instructions

As reference the Bosch HMX-celll is to be applied

The construction of a station is to be performed on the welded frames (in the event of minor loading, aluminum profile construction is also applicable). That means, each cell is an

independent machine, which is conform to the current machine guideline. The cell can be easily exchanged and is to be equipped as following:

- transport points / lashing eyes for cell transport,
- forklift feeding guide
- Painted surfaces should be defined by the location. Ex: machine RAL 7035 light gray, control cabinet / clamping, RAL 7035 light gray.

The stations must be designed in the way of a standardized uniform overall impression (structure, frame, lighting, standardized application of supply elements, protective equipment, information boards for operators etc.).

Protection doors in the opening position, must not impede the manufacturing personnel, regarding to the safety and planned manufacturing tasks. This could be achieved, for example, by means of sliding doors or folding doors opening upwards. Besides that, the doors must be reachable by both short and tall persons (for those doors opening upwards, for example, via a closing assistance). In the opening position the doors should not be in the way of passage. The design must be aligned with the RB project manager.

The station enclosure is to be made of ESD-capable Makrolon glass or ESD-capable laminated safety glass (VSG). Single-pane safety glass (ESG) is not allowed.

The company logo or signs, that leads to the identity of the manufacturer, are not allowed on delivered equipment.

If **aluminum profiles** are used for the station frame, the products of are generally to be used

Individual stations have to be **secured** together or to the floor.

All **electrical and pneumatic connections** can be realized by means of **plug connectors**.

Pinch points are to be covered. For belt inlets / outlets, protective tunnels or suitable covers are expected.

For description of the machine **functionality** a **flow chart** is expected. This is the basis for the software programming and must be signed by the SW programmer. If necessary, changes should be incorporated.

A dazzle-free illumination with consideration of the **luminance** is to be evaluated in each cell (for example Bosch system light). This lighting must be dimensioned so that processes that occur can be viewed by the personnel and possibly assessed (EN ISO 12464). In addition, energy-efficient lamps should be selected, and the lamp should be switched off (cost reduction). The switch for this purpose is to be installed outside the protective housing. In the area of optical devices or queries (light barriers, camera systems or similar) these luminaires are to be deactivated in the operating mode "Automatic" via the system control. No glossy surfaces are allowed in these areas.

Control panels should swivel, easily accessible at each manufacturing cell and not to be installed in traffic or communication routes.

Containers for product raw materials must be monitored and a signaled at low content (standard: N54 B1, coordination with RB project manager is required).

Containers for waste must be monitored and signaled at the appropriate level (standard: N54 B1, coordination with RB project manager is required).
If waste water accumulates during the process, indicate the composition and quantity.

Flexible hoses are to be secured to hose nozzles with double clamps or the hose connections must be secured against slipping off the hose nozzles.

For thermal processes (heat, cold) the pipe and container must be isolated properly.

In case that cooling water is required, it is expected that the heat exchangers with cooling tower water (plant specific, for example 26°C/36°C) to be implemented.

In general, the medium supply should be delivered according to the demand (minimum principle, demanded oriented regulation).

B.2.4.2 Accessibility and handling

A good accessibility for service, maintenance, and part placement must be ensured. The machine and equipment of Bosch must be designed in the way of “TRBS 1112 Maintenance and Repair”, so that the assigned personnel do not expose to any dangers. If the sub-assembly (i.e. cell, electric cabinets or furnace) needs to be installed higher than the Norm requirement, following attentions must be paid:

- The manufacturer is obligated to make additional risk assessment, which is required anyway in the frame of risk evaluation. The manufacturer must define the alternative or contra-measures
- If certain activities must be done regularly (maintenance or repair) at the elevated place, a proper platform should be provided by machine suppliers, which are compatible to the valid instruction. The access to the machine must be state of the art, which means, the access should be possible without additional assistance. In case of any doubt it is recommended to contact HSE department via Bosch project leader
- For operation at the elevated working places, where exists risk of falling, such area can only assessed by the scaffold, with lifting platform or ladder with railings or fall protection. For these special measures additional space must be planned in the layout. Besides that the additional cost of such measures must be estimated.

All the remaining jurist principles and specification keep to be valid.

Openings-, claddings- and caps for maintenance etc. should be guarded with disruptive safety mechanism to eliminate risk from behind. Alternatively, the danger could be eliminated by an enclosure.

All doors and walls must be able to be disassembled easily for maintenance purpose. To open them the standard tools are expected.



Picture 3 cover wall, opening with standard tools

For doors and covers, which are regularly opened, the square (mechanical) and double bit keys (electric) are required.

To be easily assemble and disassemble the machine components for maintenance and repair purpose the machine should be equipped with dedicated mechanism (i.e. quick clamping, clamping lever, pins, foldable on demand, however positional accuracy)

All function elements, like carriage guidance, need to be locked with pins.

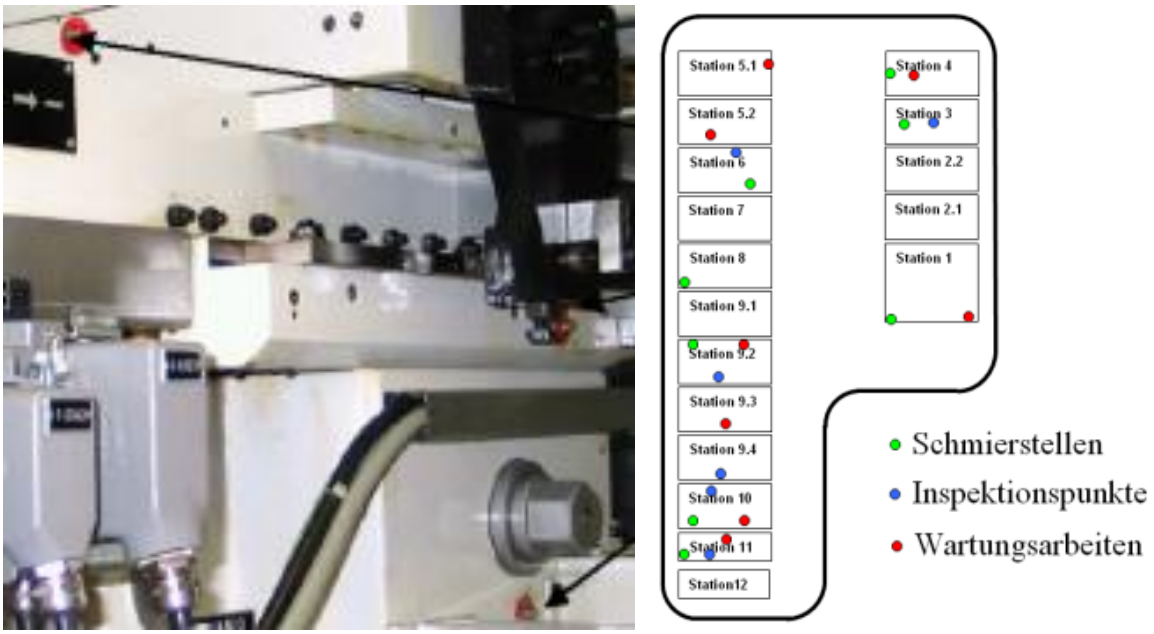
Guidance and guide rails should be accessible and can be disassembled easily. Greasing spots should be marked individually.

For rolling, chaining and roping the norm parts are expected. The application of rope is to be avoided.

B.2.4.3 Cooling and greasing materials /lubricants

The impermeability of the machine against the leakage of coolants, grease or hydraulic oil must be ensured. Only those oils and greases can be used, which are available at Bosch warehouse. The list of hazardous substances registered at Bosch can be used for hazardous substances is available from Bosch project leader. The coolants and greasing containers must be deposited centrally in the Bosch plant and easily available.

A greasing plan, in accordance to DIN 8659-2 should be developed. The greasing spots must be visible and described in the line layout. The hidden greasing spots must be pointed out separately. The greasing spots must accessable without disassembling other components.



Picture 4 marking of greasing spots and overview of line layout

B.2.4.4 Extraction facility and shop floor extraction

The extraction or suction facility for the product remnants (welding gas or particles) should be design with enough power and protected against clogging. The extraction design follows following rules:

- For extraction of hazard substances, the facility must be monitored. The faults should be displayed via signal
- The extraction and filter devices are to be marked according to their hazard assessment
- A defined measure of the extraction volume is required.
- For the extracted substance or the resultant mixture (In case of extraction from different sources with one extraction machine) the combustive number have to be evaluated.
- If the combustive number is >3 further fire preventions in the extraction machine is obligatory. Including e.g. the connection to the automatic fire alarm and the monitoring of the fire prevention.

When connected to the home exhaust system, all impurities originating from the production process must be retained in terms of plant technology. If this is not possible, appropriate measures must be coordinated with the building services operators. The compliance with the limit values of the "technical instructions air" and the "Federal emission protection law" are binding.
If the process exhaust air influences the respective processes, connection points must be provided for this purpose. If this is not possible, this should be discussed with the RB project manager

B.2.4.5 Handling with components subjected to inspection

The integration of the components, which need to be inspected, like pressure vessel, X-Ray equipment must be aligned with Bosch project leader. The proof of the inspection prior to

release may require handover protocol and/or physical marking to guarantee the inspection was performed. This also applies to the DGUV V3 exam.

B.2.4.6 Guideline for wear parts and hard stopper

Wear parts and other machine parts, which need regular maintenance, should be designed and configured in the way that they can be easily accessed and changed by operator or maintenance personnel without special tools.

Hard stopper should be made by fully hardened steel, which reaches the hardness of HRC60 – 63 as to avoidance of wear and resulted precision. In general, standardized parts are preferred and the special designed parts should be avoided where possible.

For wear parts which are subjected to strong surface loads (like gripper finger, bending jaw, fixture) a TiN-coating should be considered to reduce wear. As base material, which is to be coated with TiN for the anti-wear purpose, the tooling steel X155CrVMo12-1 is suggested. The wear part should be hardened in advance and be coated with plasma subsequently.

B.2.4.7 Manual working place and optical inspection place

In case of manual work or visual inspection places, the guidelines from B.2.8 and B.2.5.1 must be followed. Applicable regional documents should also be considered.

B.2.5 Bad parts disposition

The bad parts should be disposed in a predefined way. The bad part should not be damaged or changed during rejection. The part should remain unchanged in the event failure analysis on the parts is performed

B.2.5.1 Manual bad part rejection

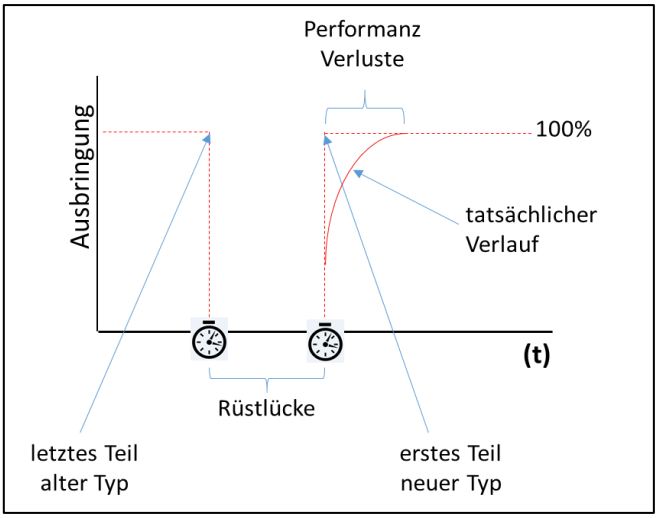
A method of securing rejected product such as a lock box should be provide where a part can be manually rejected. Access to the storage location without tool/key should not be possible.

B.2.5.2 Automatic bad part deposit

Rejected product that is automatically removed should be placed in a dedicated bin. Removal/emptying of the bin should be possible without machine stop. Tracking of the quantity of material in the bin should be monitored and require removal once a set limit is reached. The change of bin should be communicated and acknowledgement of emptying should be considered.

B.2.6 Change-over and setup for change-over parts

B.2.6.1 Nomenclates



External change-over: preparation for change-over (i.e. tools are available, parts are placed to disposal etc.).

Internal change-over: the process of change tooling hardware internal to the machine

Change-over time loss: The time measured at the end the line between the last part of the old type number and the first part of the new type number.

Performance loss: The time, which is needed, after the change-over, to reach the planned OEE

Change-over staff: The number and skill of staff needed for the internal change-over

Tools change-over: Deals with the tools requiring change manually by staff. Important is the internal change-over.

Feeding system: Deals with all feeding systems, both external and internal, which need to be emptied refilled during the changeover process, in order to produce new SNR (type number).

B.2.6.2 General requirements

Setup time optimization (short paths, same tools)

Exchange without tools. For this purpose, instead of screws, changeover friendly solutions can be found (plug-in connections, bayonet locks, clamping levers, knurled screws, star grips)

If tools are necessary, the following principles apply:

- Exchange is possible with standard tool sizes
- A common tool size must be provided throughout the system
- Tool specific labeled trays must be provided for the tools

As easy as possible

Formation of product type or setup families

Minimization of setup time (short ways, same tools)

- Exchange without tools. Therefore there must be a more setup friendly solution instead of screws (Connectors, bayonet locks, clamping levers, knurled screws, star knobs)

If tools are necessary, the following principles apply:

- Exchange is possible with standard tool sizes
- A common tool size must be provided throughout the system
- Tool specific labeled trays must be provided for the tools



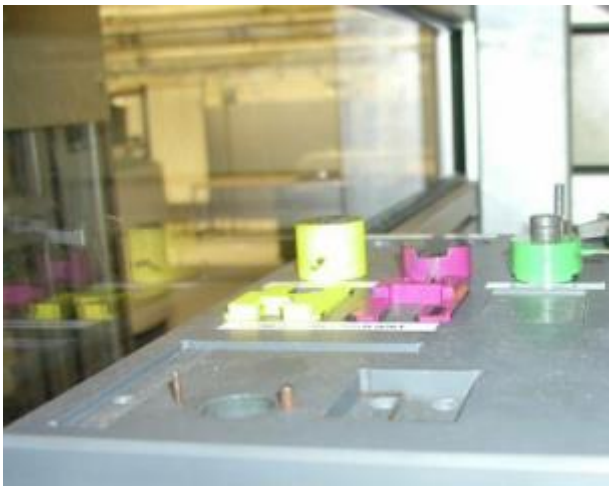
Picture 6, gripping assembly and nest assembly

- In order to avoid adjustment processes, change parts must, in principle, be designed to be quickly exchangeable with suitable positioning. This also applies to their media and energy connections. For liquid media, quick couplings with shut-off valve are to be used (for example closure nipple Z811 from Hasco), for energy connections keyed plug-in connections. Good accessibility is important.
- Defined shelves must be provided for all exchangeable parts. The following options must be agreed with the RB project manager:
  - In the ward or station-related in the immediate vicinity
  - on suitable equipment vehicles (assigned in sections for larger systems) at the central location of the production plant (interchangeable part cabinet or similar)



Picture 7 defined disposal

- Interchangeable parts have to be stored in a “confusion-proof” system ((shadow box) or mechanically coded. The design has to be agreed with the RB project manager.
- To avoid mistakes and confusion during the set-up process, in relation to interchangable parts and the position of it, the following principles apply:
  - Positioning from Interchangeable parts with own instruments (pinned fitting/ drill bushing/...) in the layer; pins have to be fixed captive.
  - Interchangeable parts have to be marked distinctive and type related (e.g. with colors)
  - Interchangeable parts have to be marked with drawing numbers (grave)



Picture 8 change part with different colors

B.2.7 ESD Safety

Applicable documents, laws and standards N51M-M24 - ESD- confirm manufacturing components

B.2.8 Ergonomics at the workplace

B.2.8.1 Valid law and norms

Other applicable documents, laws and standards
• N62A 3.3 planning criteria, ergonomic criteria, documents and pre printings
• DIN EN 1005-4 Safety of machinery - human physical performance - Part 4 Assessment of postures and movements when working on machinery.
• DIN EN ISO 14738 Anthropometric requirements for the design of machine workstations.

The timeliness of the standards is to be checked for themselves.

B.2.8.2 Occupational Regulation

The list of requirements for ergonomic workplace configuration must be filled in, together with a TEF6 working planner/operations manager.

B.2.8.3 General instruction

Already in the concept development phase, the work planner / method master of the client must be included in the design of workplaces / systems. This also applies to compliance with permissible physical employee burdens and prescribed standing-walking criteria.

When creating / modifying workplaces / systems, the list of requirements for ergonomic workplace design (see B2.8.2) must be applied. It is to be processed by the contractor and sent back with the offer. The requirement list is part of the requirements and specifications.

Unfavorable postures and hand positions, high force, as well as high-frequency movements of the upper extremities may lead to overloading and should be avoided.

The noise emissions (emission sound pressure level) shall not exceed 75 dB (A). The proof must be provided by the contractor as part of the preliminary acceptance.

B.2.9 Cleanliness requirements, cleaning

In order to avoid contamination of the products and equipment by dust from the environment, the cells and the conveyor belt have to be encapsulated (enclosure and cover).
When designing the equipment, the possibility of contamination caused by the manufacturing processes or the environment and their effect on the product must be considered and reduced as much as possible. For products with special cleanliness requirements, a cleanliness review have to be carried out after the concept phase or in the design review. For this purpose, an external company can be commissioned by Bosch for the implementation.

Due to the easy accessibility of the contamination-critical areas, an uncomplicated and effective cleaning have to be ensured. This is achieved by:

- The systems have a ground clearance (cleaning height) of at least 100 mm and all cables and lines are routed in the systems.
- All lateral and upward grooves, holes or similar closed or covered.
- The systems are either flush with each other or sufficiently spaced so that they can be easily cleaned in between.
- There is sufficient space in all areas to allow a standard vacuum cleaner to be used for cleaning.
- Contamination-critical processes are separated from the rest of the production area.
- Conveyor belts and cable towers are realized in such a way that they do not cause soiling to enter the manufacturing process.

Any production residues (eg shavings, punching residues) must be collected in separate, easily accessible containers.

Abrasion and wear of internal machine elements and the resulting partial malfunction of these can also cause contamination of product raw materials / products / workpiece fixtures. It is, for example, to check whether a waiting position of a placement gripper is required directly above a loading place, or an intermediate stop must be provided

B.2.10 Pneumatic

Other applicable documents, laws and standards

- Pneumatic equipment of machines N51M-M26

B.2.10.1 Execution details

The cross-section of the feed is to be calculated in relation to the machine consumption and the available pressure. In the calculation, a fine filter must be included. For safety reasons, the infeed has to be sized larger and has to be marked accordingly in the layout.

When using pneumatic actuators, the air losses during switching must be minimized, e.g. through the shortest possible hose lengths, small cylinder volumes or force-reduced return stroke with lower pressure levels.

When using larger actuators, the economical use of an electromechanical actuator instead of a pneumatic actuator must be checked.

The generation of vacuum preferably takes place via vacuum pumps. For small amounts of air required, energy-efficient venturi ejectors are also permissible.

The use of compressed air for blowing out or blowing off as well as blocking air is to be avoided.

For assembly lines, an air quantity meter shall be provided at the central compressed air supply. The acquisition and processing of the counter value must be defined. (for example, recording via PLC and MES or via an energy management system / the building management system).

Process-relevant traversing movements are preferably to be realized by electrical axes. Pneumatic movements for e.g. Joining and press-fitting processes are only permissible if a malfunction and a risk to the quality of the product can be ruled out through tolerance and risk assessment. Proof must be provided.

Furthermore, the following points should be noted:

- If process-relevant speeds are to be realized via pneumatic drives (eg for axies), then pneumatic fixed resistors (eg from Wiesner, selection according to the required travel speed) or adjustable precision throttles with scaling (eg Festo-GRP-70-1 / 8 -AL)

- If process-relevant forces are to be realized via pneumatic drives (for example, in the case of axes, component grippers), then precision pressure regulators are lockable (integrated lock) and manometers (for example Festo - LRPS-1 / 4-10) must be used

- If several or different forces are necessary, they can be realized via proportional valves

All devices must be uniquely marked within a system, the markings must correspond to the wiring diagram, the parts list and the drawings. Throttling check valves are generally to be installed on the exhaust airside.

B.2.10.2 Execution maintenance unit

See preferred equipment list Appendix M04.

B.2.10.3 Execution valve cluster

See preferred equipment list Appendix M04. Additionally reserve slots for valves (valve cluster<20 valve min. 2 pieces) have to be installed. At the air discharge opening, sound absorbers have to be installed (noise emission).

B.2.10.4 Vacuum

The generation of vacuum preferably takes place via vacuum pumps. For small amounts of air is required, energy-efficient Venturi ejectors are also permissible.

Instead of ejectors, the connection to the house vacuum should be provided and a line-internal vacuum pump should be used as redundancy. The exceptions are currently with a suction volume <1.9m³ / h, a duty cycle <25% of the process time or a mounting location that does not allow pumps (for example, moving axes). Vacuum filters must be easily accessible and exchangeable without special tools and without dismantling other components.

Furthermore:

- If no central vacuum supply is installed, ensure primary supply by the supplier,
- The exact place of installation and the required available power must be agreed by the RB project manager and FCM

B.2.11 Hydraulic

Other applicable documents, laws and standards

- N51M-M25 equipment of machinery

The hydraulic system must be protected against exceeding the maximum pressure (installation of a pressure switch). Use pressure-monitored centralized lubrication or maintenance-free elements. All lubrication containers must be in a central location, easily accessible and filled during operation of the system. A plate with the type and the amount of hydraulic fluid to be filled must be attached in the vicinity of the tank filling opening. Filters must be easily accessible and change without special tools or disassembly of other components.

Tank levels must be monitored electronically (hydraulics, lubrication, cooling water). The level indicators must be clearly visible and set points incl. Tolerance range must be indicated next to or on the display.

All hydraulic connections must be marked with the thread designation. A marked plate should be attached to screw-in hole or to the screw-in pin (hose).

Hydraulic cables in cable tray are to be laid individually and parallel. Separators have to be used. The bending radius must be observed and they must be intercepted on both sides.

The hydraulic supply must be designed to be energy-efficient. Pressure control is preferable via speed control of the pumps (pressure control valves are to be avoided). For small flow rates, the use of a storage charging circuit should be checked (intermittent operation of the pump).

When using larger actuators, the economical use of an electro actuator instead of a hydraulic actuator has to be checked.

If higher hydraulic pressures are required, the use of a pressure booster must be checked

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B.3 Electrics and software engineering

B.3.1 Electric design instruction

Further applicable documents, regulation und Norms

- Bosch-Norm N54 B1
- Bosch-Norm N54 B2
- Bosch-Norm N54 B3
- Bosch-Norm N54 D2
- Bosch Norm N54 D4
- Bosch-Norm N54 D6

Electric drive systems are to be designed energy-efficient:

- Use of electric motors of efficiency class> = IE3
- Operation of the electric motors close to the rated working point (P / Pnenn> 0.8)
- Use of speed control when economical
- Regenerative braking energy if economical (eg DC bus coupling in multi-axis systems)

For machine lighting in continuous operation (manual workstations), illuminants of efficiency class A or better, preferably LED.

Cabinets are preferably designed so that no or only passive cooling is necessary. If required for active cooling, a water-cooled cooling unit should be provided. Otherwise, particularly energy-efficient air-conditioned air conditioning units are to be provided.

For assembly lines, an electrical energy meter (to be defined as the preferred device) must be provided at the central power supply. The acquisition and processing of the counter value must be defined. (for example, recording via PLC and MES or via an energy management system / the building management system).

B.3.1.1 Tooling

The schematic documentation must be created with the following tool.

Tool: EPLAN Electric P8

Version: 2.8

B.3.1.2 Pattern plan

In a very early phase of the project and before the start of the electrical construction, it must be clarified with the client whether and with which sample plan the electrical construction must be carried out.
The electrical design must be carried out on the basis of the model plan of NEXEED Automation by BOSCH BCI and use the parameters and settings specified therein.

B.3.1.3 Construction guidelines, BMK guidelines

When constructing the electrical construction, the BOSCH construction guidelines must be applied and followed.
When creating the electrical construction, the BOSCH guidelines must be applied and followed.

B.3.1.4 Fieldbus systems

EtherCAT is used for NEXEED automation systems.
Possibly. Additionally, required field bus system must be discharged as sub-bus from the EtherCAT.

B.3.1.5 Competence Management

The construction of the electrical construction must be carried out exclusively by specially trained designers.
This means that the electrical design is carried out or accompanied by designers who have completed the appropriate training of BOSCH BCI.
For projects with more than one single station, the contractor must designate a lead designer who is responsible for compliance with the design guidelines and specifications of the client.
The lead designer must have the necessary experience and skills.

B.3.1.6 Hardware devices

For the creation of the electrical construction, the hardware devices agreed with the client must be used.

In a very early phase of the project, a delta analysis must be carried out to what extent the devices installed in the electrical design are represented and covered by a software object in the NEXEED Automation object repository and by a device EPLAN macro for the electrical design.
Possibly missing software components for individual devices must be identified as quickly as possible and their creation or procurement ensured.

B.3.1.7 Device arrangement

A maintenance & service plug socket, which is still a live power source while the main switch is off, should be planned in the control cabinet. Further to be planned are a plug socket, which is a live power source only while the main switch is on, and a network connection RJ45 for service purpose. They should be installed at the main operation side of each station, which are easy accessible (attention to protection class).

All plug sockets, which are accessible to electric layman, should be secured by the FI-protection switch IΔn ≤ 30mA. This rule is also valid for plug sockets, which can be accessible through the protection facilities.

The display and control unit of the station control, must be placed in a high of 160cm (mid of display).

B.3.1.8 Color of press buttons, control elements and indicator lamps (bollard)

Colors according to Bosch Norm N54 B1 general specification special machine, special equipment and special mechanism 8.4. The colors are ruled in the Bosch Norm N54 B1.

B.3.2 Interfaces and connections

If the machine use OPCON, communication must be made by an XML frame. If it's not, (Siemens for example), we must have an OPC-UA server.

B.3.2.1 MES connection

Machine can be either connected to OpCon MES or ProMaster via one of following three communication channels. Please chose it.

OpCon MES		☒	☐
1.	DDL via XML frames		☒
2.	Direct data exchange OPC-DA		☐
3.	OPC-UA		☒
4.	without		☐



When designing a station, the communication time for data exchange with the MES must be taken into account. The communication time must be taken into account when calculating the cycle time.

B.3.2.2 M2M connection

- For direct communication between machines based on the automation system NEXEED Automation, the "DataBridge" from NEXEED Automation from BOSCH BCI must be used.
- If real-time communication is required (between two machines), the use of a "bus bridge" (for example EtherCAT Bridge) is preferred.
- Network variables

B.3.3 Control technology / automation software

Preferential Control Software is NEXEED Automation from BOSCH BCI

It must be ensured that the machine can be easily switched off and put back into operation by the operator.

If a "warm-up" of the machine is necessary, it must be carried out in such a way that the machine can be used again as quickly as possible (for example, avoiding parallel heating and cooling during warm-up).

Auxiliaries and larger consumers (for example, media supply, extraction systems) are only switched on in automatic mode when required and are otherwise switched off or reduced during operation.

The machine supports a Standby mode in which auxiliary units and power units are switched off or reduced as far as possible. It activates this state on request and, if necessary, independently after a configurable period of inactivity (partial deficiency, malfunction).

Machine lights, which are only needed if required (interior, automatic stations) must be switched off in normal operation and only switched on for a configurable time on request.

B.3.3.1 NEXEED Automation –Control plus

The automation software must be created by using NEXEED Automation.

The latest version of NEXEED Automation have to be used at the start of the project, unless the client does not stipulate it differently.

During implementation, the NEXEED Automation Coding Guideline must be applied and followed. The same functions and applications must also have the same name and handling across stations.

The function / sequence of the station is to be displayed via a schedule (including MES communication) and stored in the directory "<Station> \ Docu".

It must also be ensured that any HMI style guide that is valid for the project is taken into account when implementing the user interface. The same controls and display windows must have the same look across stations.

During the project phase the structure of software and knowhow must be ensured. Option is subscribing to and considering the NEXEED Automation Newsletter.

The hardware components published and released in the NEXEED Automation product catalog are available as control hardware. From this hardware portfolio, the controller hardware have to be selected.

While implementing the control software, strict adherence to the BOSCH BMK equipment labeling have to be ensured, which must already be used in electrical engineering.

B.3.3.2 NEXEED Automation objects

To create the automation software with NEXEED Automation, use the software objects from the NEXEED Automation Object Repository.
In a very early project phase, a delta analysis must be carried out to determine to what extent the software objects contained in the NEXEED Automation object repository cover the project requirements. Possibly missing software modules for individual fieldbus devices have to be identified as quickly as possible and their creation or procurement ensured.
BOSCH BCI must be integrated at an early stage, both when considering the delta and when creating or procuring missing software components.

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B.3.3.3 Competence management

The automation software with NEXEED Automation must only be executed by trained and certified programmers.

This means that any programmer who creates the automation software with NEXEED Automation has attended the basic training OP120 and can prove this with a certificate.

For projects with more than one single station, the contractor must name a lead programmer who is responsible for compliance with programming standards, interface definitions and client specifications. The lead programmer must have the necessary experience and skills.

In the case of own software object creation by the contractor, it must be ensured that the executing programmers have the necessary qualifications and have visited the object training OP150 required for this purpose.

B.3.3.4 Basis project

In a very early phase of the project and before the start of the software development, the client has to clarify whether and with what basic project the software development has to be carried out.

B.3.3.5 Consulting

Depending on the experience and expertise of the contractor, it makes sense to use project consulting from BOSCH BCI.

It's the task of the contractor to identify his own needs for project consulting at an early stage and to contact BOSCH BCI in order to ensure the necessary support during the project.
Possible project consulting offers are:

- Feasibility analyzes
- Assistance in the preparation of your technical concept and specifications
- Holistic project planning and documentation
- Project management and coaching
- Configuration including PLC programming, visualization, network technology etc.
- Support during commissioning on site
- Accompaniment and optimization in the current production plant

B.3.3.6 MES connection

In a very early project phase, the coordination with the MES application has to be carried out and it must be ensured that both communication technology and communication content are taken into account and implemented in the automation software.
The contractor must implement the requirements and specifications for the data interface in accordance with the MES specification.

B.3.3.7 Functions and processes

In a very early project phase, special functions and processes have to be detected and queried as to whether the necessary support is available in NEXEED Automation. This study could be performed by using BOSCH BCI.

Special functions and processes are e.g. things like:

- Lab-View connection
- Matlab connection
- M2M connections
- connect IT systems
-

B.3.3.8 Software backup and configuration management

The use of configuration management and the safeguarding of software baselines is an elementary prerequisite for successful project execution.
The contractor must consistently apply the software management specified by the client and consistently apply the processes and processes prescribed by the client in configuration and baseline management

B.3.3.9 Data security

The protection of personal data and compliance with data protection regulations are very important to Bosch.
As a developer of automation software, the contractor must take the requirements of the General Data Protection Regulation (DSGVO) seriously and implement these in its products by providing technical options for implementing data protection regulations.

B.3.3.10 Data safety technology

The interface of the PLC to the safety technology must be based on the standard specification. The definition could be found in standard N54.

B.3.3.11 Software acceptance

At the conclusion of the project, the contractor must prove with a control plus software acceptance checklist that all prescribed guidelines and specifications have been complied with. The Bosch project manager provides the software acceptance checklist to the contractor and this represents the minimum scope.

B.4 Application mechantronic components

B.4.1 Robot application

For 6 axis application Kuka is the preferred robot. For the Scara application Stäubli is the preferred robot

B.4.1.1 Robot application

According to DIN EN ISO 10218-1 - industrial robots

B.4.1.2 Hardware requirements

KUKA
Robot: 6-axis Agilus series robots (preferably)
Robot control: Agilus with KRC4 Compact control

All 6-axis robots of the KUKA Agilus series may be used, including Agilus
System extensions such as an additional linear axis system.

In consultation with the ordering department, KUKA robots, which have the same robot control (KRC4) and are identical in robot programming, can also be used. Reason: e.g. with a larger radius of action of the robot.

Staubli robot

Robots: SCARA TS60 (preferably), TS40, TS80

Fast Picker TP80

6-axis robot TX40, TX60 (preferably), TX90 also L / XL versions

Robot control: CS8C (Coming soon CS9)

B.4.1.3 Basis software

A command-based robot base software is provided as a template for a KUKA or Stäubli robot and should be used. The robot is integrated as a slave in a PLC system.

This basic robot software serves as a template, but does not contain a machine-specific, executable program. The basic structure of this basic software must be retained and includes e.g. following features:

- cell configuration (sample application)
- Interface to OPCON
- Structure for PLC control of the automatic process
- Reset / Homing functionality
- grab control
- Output error messages / error numbers

The responsibility for an executable robot program lies with the robot
Programmer and not at the ordering department.

B.4.1.4 Protection equipment

The protective devices of the robot manufacturers must be used and integrated in the protective device of the machine. The protective enclosure must be installed so generously that the robot, including attachments, does not reach it. If this is structurally not possible, reinforced protection must be fitted and appropriate door bolts used.

B.4.1.5 Documentation

If necessary, suitable setup aids / teach aids should be provided for the teach points.

In a documentation, it must be noted which teach points are taught with which tool and possibly with which teaching aid. This documentation will be provided at the latest with the device software

B.4.2 Guideline for automatic placement and palletizing

The feasibility and capability of the packaging design and its characteristics (palletizer, blister, box) should be examined and verified. Following influence factors must be considered:

- Position tolerance (i.e. safe stand, pack height, material stability, bending, deformation)
- Anti-twist safety (i.e. strengthening features, cut, bolt pin, holes are available, outer contour clearly defined)

Packaging usually has higher tolerances. For this reason, a tolerance consideration for the entire process must be carried out to ensure the removal and placement accuracy.

It is also important to pay attention to what momentary state the packaging can have during processing by the system and how the system can influence the condition of the packaging (temperature, humidity, cleanliness, damage).

Concerns about the planned packaging or the process must be agreed with the RB project manager at an early stage.

B.4.3 Separation and feeding system

Because of flexibility and later more reusing reasons, Any-Feeder off 2 part variations and suitable geometry are required. If this is not possible, because of technical reasons, it has to be clarified with the Project manager.

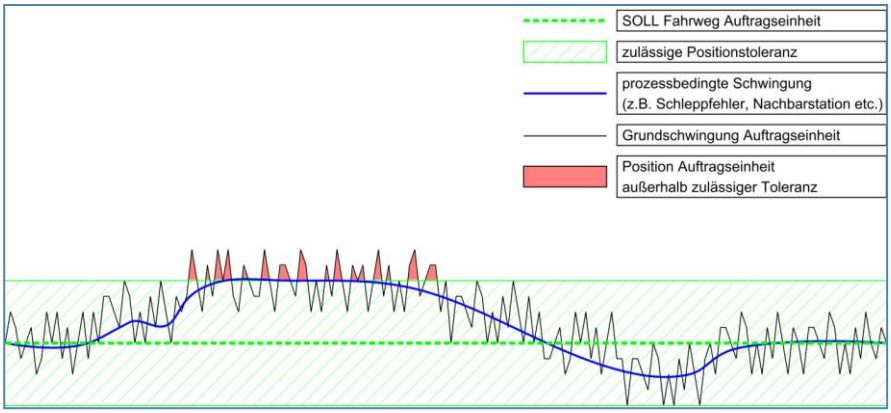
B.4.4 Application of axle system

It is constructive to provide a defined fixed point (coordinate origin) for the calibration of axes and position sensors. The position of the joining partner in the machining position must be ensured by suitable design measures or detected by position detection. Referencing of the plant is to the center of the active part of the process module, e.g. Outlet opening of the dispensing needle or center of the welding electrode etc.

Determined positional deviations of joint partner and / or process module are to be offset and tracked at runtime. In the case of direct measurement, the relevant external contours of the

active part of the process module must be completely recorded in the relevant spatial directions (± X, ± Y, ± Z, ± F, etc.).

Movements of the job unit and / or neighboring machining stations must not cause the job unit to swing outside the required tolerances at the target cycle (see Figure 2).



Positioning accuracies must be verified for the entire system, including the auxiliary processes (referencing, position detection of the joining partner, etc.). Compliance with the required positioning tolerances during the machining process including the vibrations of the

The order unit shall be evidenced by the system manufacturer at the installation site at the target location by means of suitable acceptance procedures.

B.4.5 Clamping system

Clamping systems are preferably operated by spring force, the relaxation can then be done pneumatically or hydraulically.

B.4.6 Transfer system

The systems of the Bosch Rexroth TS series have to be used as a transfer. The transfer system must be separable by cells (aluminum profile from the belt system). The conveyor belt may be driven by several cells throughout and with a motor. The tape length is to be interpreted according to the load. All stoppers and lifters must be assigned to the respective station and must be marked captive. WT lifting stations in a band system must be installed and pinned in such a way that the positioning accuracy of the processing station is always met. The belt motors must be installed in such a way that enough space is available for the replacement / disassembly of the motors. Tape systems are to be covered from the lay-up to the packaging place. Execution and, if necessary, deviations must be agreed with the purchaser.

The work piece carrier design is to be presented to the client with a sample work piece carrier for the design review. The client releases the work piece carrier production.

The work piece carriers must be marked captive with the following information (engraved or engraved plate screwed on):

- Drawing number,
- Consecutive WT no. with line assignment.

If it is not possible to rule out falling down of the work piece carriers due to top load or protruding parts, these must be secured by suitable measures. e.g. additional leadership or attack.

The identification system must be used in accordance with Appendix M07.

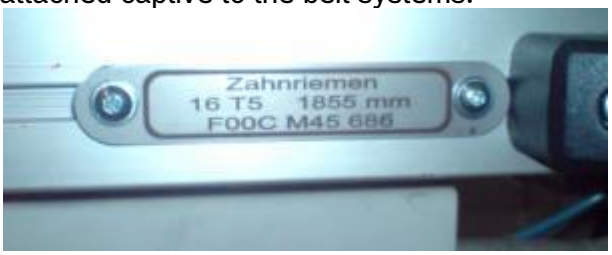
For the belt section standard assemblies (endless belt belt) are to be used. Expensive special solutions (chains or similar) should be avoided.

At locks within the transfer, redundant stoppers should be used to prevent the work piece carriers from falling down.

There must be openings / free spaces for belt tensioning, cleaning and maintenance



By using of toothed belt, only the use of normed sizes is allowed. Belt length and type are to be attached captive to the belt systems.



Empty and full belt sections are automatically switched off with automatic restart.
Energy-efficient belt drives (motor / gearbox combination) are to be used

- The transfer system can be implemented via a separate control system independent of the station controls.
- For loosely linked systems using a transfer system, this must be arranged electrically in such a way that largely continuous strip sections with one drive are created. An electrical separation of the band sections per station is not desired.

B.5 Inventory of IT devices and emergency concept

B.5.1 Inventory

IT devices that fulfill at least one of the following characteristics must be inventoried:

- The IT terminal is assigned an IP address
- The IT terminal can be reached via a network
- The IT terminal processes data that can be exchanged via an accessible data interface (for example Bluetooth, Ethernet, mobile mass storage such as USB stick, SD card or CD / DVD drive, serial, etc.)

Examples: computers, controllers, active display panels, cameras, robot controls, process controls, welding controls, screwdrivers etc.

The following data must be documented:

- IP address
- MAC address (optional)
- Computer name
- Serial number
- Operating system

The component level (actuators and sensors) is excluded.

In order to know the used IT devices in one machine, it is important that the following documents are included:

Delivery instructions for machines

Delivery instructions for machinery

N51M_M27_2015-08-14_Appendix_IT_Sicherheitsfragebogen_für_MAE.xlsx

In it the used IT terminals, user identifications etc. must be listed.

The overview of the IT devices for inventory can also be used via the Bosch IP address list template.

All installed IT devices must be provided with a computer number sticker.

B.5.2 IT- emergency concept

The IT emergency concept describes the data backup and the restoration of the software of the IT components installed in the MAE. It also describes all the media required for recovery, connectivity technologies, development tools, and prerequisites for use. Part of the emergency concept, however, is the prior backup and provision of the software.

The core is the backup and recovery of the operating system and the application programs, in order to get the machine back to production capacity in the event of a system failure. IT contingency plan:

The specific IT contingency plan is dedicated to recovering the operating system and software needed to operate the MAE in all installed IT components, as well as their subsystems (for example, using a separate process control subordinate to the PLC). The IT contingency plan may take the form of checklists, depending on the particular

Event to be created. Events in which an emergency plan is used include, for example, power failures, technical malfunctions, hacker attacks, virus attacks, staff shortages or operating errors. Here are the key contents of an IT contingency plan to deliver:

- Directory of all documentation and information relevant to emergencies
- Checklists with recommended actions
- Checklists for error limitation and problem analysis
- Possible workarounds and emergency operation procedures
- Procedure to restore the function
- Contact lists and responsibilities at the MAE supplier
- Documentation of emergency access data and access information
- Residual risks and service concepts

B.6 Remote maintenance service

remote access refers to the spatially separate access to the machine by a person (technician internal or external, production planner, TEF), which does not physically act on the machine.

Is remote access planned for the machine?

☐ NO Access is only possible physically, directly on the machine.

☐ YES Access is only possible via Bosch's own remote maintenance solution RSA

(Remote Shop floor Access) possible. Access via remote maintenance solution of the manufacturer is not permitted.

B.6.1 Information about RSA

Bosch externally:
https://sites.inside-share.bosch.com/sites/030936/Documents/MAE%20supplier

Bosch extern:
https://www.bosch-connected-industry.com/connected-manufacturing/nexeed-remote-shopfloor-access/

Needed components of the Bosch own remote maintenance solution existing mainly out of:

- RSA Router (obligatory)
- Key-operated switch or function-equivalent technology (obligatory)
- Status light (optional, existing signal lamps can be used for this)

B.6.2 Supplementary remote maintenance regulations and contracts

Documents (such as commitments and conditions and use for external service companies) are stored as templates for RB employees and are provided to the MAE client as a basis for negotiation.

https://inside-docupedia.bosch.com/confluence/display/RSA/Remote+Shopfloor+Access

B.6.3 Risk reduction measures

In the risk assessment by the remote maintenance, in accordance with the Machinery Directive and standards, therefore, the remote maintenance operation must be subjected to a special consideration. This results in increased duty of care, control and documentation for both the local operator and the remote operator.

- By using suitable safety technology, it must be ensured that the control "On" can only be switched on when protective devices are activated and can not be switched on remotely (for example by linking: pushbutton 'control on' + protective doors closed + main air OK + ...
- The operator on site may only trigger individual functions in manual mode and automatic sequences. That This must not be done during remote maintenance by the remote maintenance engineer by triggering a software function. Locking of the activated or activating remote maintenance hardware is to be provided and evaluated in the machine safety risk assessment

- The remote maintenance process itself can only be initiated by the local operator and can be interrupted by him at any time. The activation of the remote maintenance access must be protected against unauthorized activation. (e.g., key switch, password protection).

The activation of the remote maintenance access must be protected against unauthorized activation (e.g. key-operated switch, password protection).

An active remote maintenance access must be visible to the operator of the MAE (e.g. signal light).

By parameterizing the communication connection, it must be ensured that only the IT terminal to be serviced is accessible, this applies to both the IP address and the port enable.

B.7 Quality requirement

B.7.1 Applicable document

- Robert Bosch Schriftenreihe Heft 8, Messunsicherheit
- Robert Bosch Schriftenreihe Heft 9, Maschinen- und Prozessfähigkeit
- Robert Bosch Schriftenreihe Heft 10, Fähigkeit von Mess- und Prüfprozessen
- Robert Bosch Schriftenreihe Heft 14, Fehler-Möglichkeits- und Einfluss-Analyse (FMEA)

B.7.2 Investigation of machine capability (short time process capability)

The machine capability study is a short-term investigation aimed at revealing only machine-related influences on the manufacturing process. The machine capability must be proven by the manufacturer prior to delivery of the machine. The machine capability indices consider the spread and location of the process.

The detailed process description can be found in the Robert Bosch Publication Series, Book 9. Further regulations on the calculation methods can be found in ISO 22514-2.

B.7.3 Investigation of the capability of the measurement and testing system

The proof of the capability and the monitoring of the stability of measurement processes should ensure that a measuring device can measure a quality feature at the place of use with a sufficiently low systematic measurement deviation and measured value dispersion (in each case related to the feature tolerance). The procedures available for continuous (variable) features and methods for assessing discrete (attributive) characterization processes are described in the Robert Bosch Series 10 Publication.

B.7.4 Further general requirement

The response of the test device when the limit value is exceeded, must be ensured. Positioning and setting aids / gauges are made and delivered for the inspection. The client provides the required border and error patterns. However, the contractor must notify the necessity in good time.

All physical parameters to be measured (pressure, flow, current, voltage, speed, etc.) must have test connections. The connections should be as close as possible to the sensor. Type and design must be agreed with the RB project manager.

The resolution of the meters must be £ 5% of the feature tolerance. The calibration uncertainty must be <10% of the tolerance to reach Cgk≥1.33.

For each integrated measuring device, a calibration option must be provided.

B.7.5 References

Measuring instruments are to be calibrated before the quality inspection. The calibration should be done with a calibration standard or in individual cases with a master unit (serial part for calibration purposes). The nominal value must be set within the range of application of the measuring instrument, if possible to the mean value. With adjustable display, at least three measuring points in the application area (bottom, middle, top) are required. For non-work piece-related measuring equipment (universal measuring instruments), the manufacturer must provide the following information:

- Resolution,
- Repeatability with probability of statement,
- Linearity,
- Accuracy,
- Ambient conditions for this information,
- Measurement uncertainty.

This information must be documented based on a test or calibration certificate.

The detailed procedure for determining a measurement uncertainty can be found in the Robert Bosch Series 8 or GUM or VDA Volume 5.

B.7.6 FMEA

FEMA is an analytical tool of preventive quality assurance. It serves to investigate the potential weakness (for example in the manufacturing process, at tooling, machine or equipment), to evaluate and define the proper measures, in order to avoid the mistakes. The early elimination of weak points leads to the reduction of risks and costs caused by defects, through which the safety, reliability and custom satisfactory can be increased.

The procedure and methodic for FMEA is described in the booklet 14 and VDA Band 4.

B.8 Occupational safety, fire control and environment protection

Further applicable documents, regulation und Norms

N93 I3.9.4 adjuster key switch

B.8.1 Use of hazardous substances

All safety data sheets of the task-specific chemical agents, in particular also the auxiliaries, lubricants and use materials are to be handed over to the contact person electronically (PDF) to..P / HSE before use in the factory in current version.

If dangerous substances are used as intended in the production process of the MAE to be constructed, it must be ensured that they cannot be unintentionally released in all operating states according to the state of the art. If media have to be removed, they must be prepared before transport (for example, in ventilation and air conditioning systems) in such a way that there is no risk of endangering the system, the building or the environment. Deposits and condensates outside of the erected MAE must not occur. Robust proofs must be presented before acceptance.

Contamination of service personnel is to be avoided by foreseeable operating conditions through technical measures. If release is technically unavoidable, occupational exposure limits must be demonstrably safely adhered to. It is essential to consider the future installation site.

B.8.2 Marking of risks and dangers

Media lines must be marked in accordance with DIN2403 in the latest valid version at the beginning of the line, at component feedthroughs (on both sides) and at the end of the line with the flow direction.

If waste collection containers are installed, they must be marked after consulting ..P / HSE.

B.8.3 General information

Electromagnetic fields must not exceed the limit values for exposure range 2 (BGV B11). If fire protection, radiation protection, EMF and / or environmental protection measures are required due to the process or the substances used, these must be agreed with ..P / HSE.

Knowledge of the requirements of local laws and licenses available to EDL must be provided in writing to the operator at the earliest possible date. Required measures are to be coordinated with the operator immediately and ..P / HSE.

B.8.4 Special protection against fire

In the case of MAE with combustible media (e.g. dusts, KSS oil, white spirit), an explosion protection concept or fire protection concept with switch-off matrix must be presented in the design phase. The factory safety engineer (HSE) is to be involved in their approval.

The shutdown matrix serves as the basis for the decrease of the MAE.

- The following special cases also require individual clarification (without claim to completeness):
- Use or development of hazardous substances, radioactive substances (Bosch standard N 2580)
- Use of water-polluting substances
- Generation of waste or wastewater
- Emissions to the air (vapors, fumes, fog)
- Application or formation of ionizing or non-ionizing radiation (laser, X-ray, UV)

Smoke, gases and vapors which are to be removed from the MAE via the central exhaust air system of the factory must be cleaned in such a way that no deposits can form inside the system. In the case of flammable or potentially explosive mixtures, the specification of the exhaust system must be taken into account so that fire and explosion hazards can be excluded in all operating conditions or when the exhaust air system is restarted.

For automatic processes intended for mass production, the following special provisions apply to extraction at stations where steel alloys are laser-welded under protective gas:

- From the beginning of series production until the verification of the burning point of the extracted residues, an extinguishing system must be integrated into the relevant extraction device and connected to the fire alarm system
- If the number of burned residues is permanently > 3 after repeated analysis, an internally monitored neutralization of the residues with opto-acoustic indication of a malfunction must be provided
- In order to enable a subsequent reuse of the extinguishing system, a specification in accordance with the attached ..P system should be provided

B.8.5 Working execution

In the case of work carried out on our factory premises, we attach great importance to mutual guarantee of occupational safety and compliance with legal regulations. Therefore, prior to each use in our company, a coordination with the respective client is required taking into account the site-specific requirements.

Here are in particular

- Trial operation without complete function of the protective devices,

- The coordinators of the contracting entity and the contractor,
- Special approval of hazardous work (eg welding, use of lifting equipment) must be obtained.

B.9 Documentation

The Documentation is to be done according to BOSCH-Norm N51M M20, point 10 „Documentation“.

The documentation comprises of machine software, incl. user interface and possibly existing software application.

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B.10 Further norms and binding guidelines

Dokument-Nr.	Name
	The Supplementary Terms and Conditions of Purchase for Software of Robert Bosch GmbH so-called Free Open Source Software (FOSS) are part of the contract. The current version of our conditions can be found on the Internet at Delivery instructions for machinery
N51M M20	General delivery instructions for machines, plants and production facilities
N51M M23	Delivery instructions for electrical equipment of machines, plants and production facilities
N51M M24	Delivery instructions for ESD-compliant production components
N51M M27	Delivery instructions for IT security of machines, systems and production facilities (MAE)
N54 B1	General requirement on special machines, equipment and facilities
N54 B2	Control technology, electrical electronics; Project-related specifications for electrical equipment from MAE
N54 B3	Control technology, electrical electronics; Device preferred list
N54 C1	Control technology, electrical electronics; Initial acceptance of fixed machinery, equipment and facilities

N55 D2	Control technology, electrical electronics; Electrical equipment of machines, installations and manufacturing equipment; safety circuits
N54 D4	Control technology, electrical electronics; BUS systems
N54 D6	Control technology, electrical electronics; E-CAD; Design and naming rules
N55 D6	ESD protection; testing in the EPA
Heft 8	Series quality management in Bosch-Group, Technical statistic, Heft Nr. 8, measurement uncertainty
Heft 9	Series quality management in Bosch-Group, Technical statistic, Heft Nr. 9, machine and process capability
Heft 10	Series quality management in Bosch-Group, Technical statistic, Heft Nr. 10, capability of measurement and testing process
TPM Säule 4	MAE target specification